SOLUTION
SECTION-I

Single Correct Answer Type **11 Q. [3 M (–1)]**

1. Ans. (B)

Released energy = $2 \times 4 \times 7 - 2 \times 1 - 7 \times 5.4 = 16$ MeV

2. Ans. (A)

Sol. Equation of β^+-decay of

${}_6C^{11}$; ${}_6C^{11} \longrightarrow {}_5B^{11} + {}_{+1}\beta^0 + \nu + Q$

Q-value of reaction = Δmc^2

$= \left[m\left({}_6C^{11}\right) - 6m_e - m\left({}_5B^{11}\right) + 5m_e - m_e\right]c^2$

$= \left[m\left({}_6C^{11}\right) - m\left({}_5B^{11}\right) - 2m_e\right]c^2$

$= [11.011434 - 11.009305 - 2\times 0.000548]uc^2 = [0.001033]\ uc^2 = 0.001033 \times 931.5$ MeV

$= 0.962$ MeV

3. Ans. (A)

Sol. Let initially the number of nuclei of ${}^{32}_{15}P$ & ${}^{33}_{15}P$ are N_1 & N_2 respectively

so $\frac{R_1}{R_2} = \frac{\lambda_1 N_1}{\lambda_2 N_2} = \frac{90}{10}$ $(\because 90\%$ from ${}^{32}_{15}P$ & 10% from ${}^{33}_{15}P)$

Now let at time 't' 90% decays come from ${}^{33}_{15}P$ then ratio of activity.

$$\frac{R_1'}{R_2'} = \frac{\lambda_1 N_1'}{\lambda_2 N_2'} \Rightarrow \frac{R_1'}{R_2'} = \frac{\lambda_1 N_1 e^{-\lambda_1 t}}{\lambda_2 N_2 e^{-\lambda_2 t}} = \frac{10}{90} \Rightarrow \frac{90e^{-\lambda_1 t}}{10\, e^{-\lambda_2 t}} = \frac{10}{90}$$

$$\Rightarrow e^{(\lambda_1-\lambda_2)t} = 81 \Rightarrow \ell n e^{(\lambda_1-\lambda_2)t} = \ell n 3^4 \Rightarrow (\lambda_1 - \lambda_2)t = 4\ell n 3 \Rightarrow t = \frac{4\ell n 3}{\ell n 2\left(\frac{1}{T_1} - \frac{1}{T_2}\right)}$$

4. Ans. (C)

Sol. $\lambda_\alpha = \frac{1}{1620}$ per year and $\lambda_\beta = \frac{1}{405}$ per year and it is given that the fraction of the remained activity $\frac{A}{A_0}$

$= \frac{1}{4}$

Total decay constant

$$\lambda = \lambda_\alpha + \lambda_\beta = \frac{1}{1620} + \frac{1}{405} = \frac{1}{324} \text{ per year}$$

We know that $A = A_0 e^{-\lambda t} \Rightarrow t = \frac{1}{\lambda}\log_e \frac{A_0}{A}$

$$\Rightarrow t = \frac{1}{\lambda}\log_e 4 = \frac{2}{\lambda}\log_e 2$$

$= 324 \times 2 \times 0.693 = 449$ years

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5. Ans. (B)

Sol. $6.9\ \% = \frac{6.9}{100} = \frac{1}{14.5} \approx \frac{1}{16} \approx \frac{1}{2^4}$

Total time is little less than $4t_{1/2} = 4 \times 12.3$

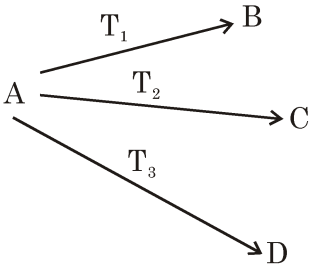
$= 49.2$

So answer T slightly less than $4t_{1/2}$

$T = 47.5$

6. Ans. (B)

Sol.



Rate is maximum is case of T_3.

So almost all nuclie decay in process $A \rightarrow D$

7. Ans. (B)

Sol. x $\longrightarrow$ y

t = 0	16% = N_0	0
t = t2% = N	14%	

$N = \frac{N_0}{8} \Rightarrow \frac{1}{8} = \frac{1}{2^n} \Rightarrow n = 3$

$t = 3T_{1/2} = 135$ yrs.

8. Ans. (C)

Sol. Given $\frac{N_0\left(1-2^{-2/t_{1/2}}\right)}{N_0\left(1-2^{-4/t_{1/2}}\right)} = \frac{n}{(0.25+1)n}$

{No. of β-particles emitted = No. of nuclei decayed} $\Rightarrow t_{1/2} = 1$ sec

9. Ans. (A)

Sol. Equation of β^+-decay of

${}_6C^{11}$; ${}_6C^{11} \longrightarrow {}_5B^{11} + {}_{+1}\beta^0 + \nu + Q$

Q-value of reaction = Δmc^2

$= \left[m\left({}_6C^{11}\right) - 6m_e - m\left({}_5B^{11}\right) + 5m_e - m_e\right]c^2$

$= \left[m\left({}_6C^{11}\right) - m\left({}_5B^{11}\right) - 2m_e\right]c^2$

$= [11.011434 - 11.009305 - 2 \times 0.000548]uc^2 = [0.001033]\ uc^2 = 0.001033 \times 931.5$ MeV

$= 0.962$ MeV

10. Ans. (A)

Sol. The rate of productive of the element A is α while its rate of decay is $(-\lambda N)$. Thus the net rate of change of nuclei A is given by $\frac{dN}{dt} = (\alpha = -\lambda N)$; $\frac{dN}{(\alpha - \lambda N)} = dt$

Integration $\int_{N_0}^{N} \frac{dN}{\alpha - \lambda N} = \int_0^t dt \ \ -\frac{1}{\lambda}\left[\left|\log_e(\alpha - \lambda N)\right|_{N_0}^{N}\right] = t$; $\alpha - \lambda N = (\alpha - N_0)e^{-\lambda t}$

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Which gives the number of nuclei at time t : $N = \frac{1}{\lambda}\left[\alpha - (\alpha - \lambda N_0)e^{-\lambda t}\right]$

11. Ans. (C)

Sol. Electrostatic energy = $BE_N - BE_O$

$= \left[[7M_H + 8M_n - M_N] - [8M_H + 7M_n - M_O]\right] \times C^2$

$= [-M_H + M_n + M_O - M_N]C^2$

$= [-1.007825 + 1.008665 + 15.003065 - 15.000109] \times 931.5$

$= + 3.5359$ MeV

$$\Delta E = \frac{3}{5} \times \frac{1.44 \times 8 \times 7}{R} - \frac{3}{5} \times \frac{1.44 \times 7 \times 6}{R} = 3.5359$$

$$R = \frac{3 \times 1.44 \times 14}{5 \times 3.5359} = 3.42\text{fm}$$

Multiple Correct Answer Type — 6 Q. [4 M (–1)]

12. Ans. (A, B, D)

Sol. $R = R_0A^{1/3}$

$A^{1/3} = 6$

$A_P = 216$

for daughter nucleus

$A_d = 212$

Also $\frac{212 - Z_d}{Z_d} = \frac{65}{41}$

$(41)(212) = 106\, Z_d$

$Z_d = 82$

$Z_P = 84$

13. Ans. (B, C)

Sol. $A = A_0e^{-\lambda t}$

$6.932 \times 10^{18} = A_0e^{-\frac{\ell n 2}{40}(80)}$

$= A_0e^{-\ell n 4} = \frac{A_0}{4}$

$A_0 = 4 \times 6.932 \times 10^{18}$

$\lambda N_0 = 4 \times 6.932 \times 10^{18}$

$\frac{0.6932}{40} N_0 = 4 \times 6.932 \times 10^{18}$

$N_0 = 160 \times 10^{19}$

$N_0 = 1.6 \times 10^{21}$

Also $\frac{3}{4}(1.6 \times 10^{21})E_0 = 6 \times 10^8$

$E_0 = 5 \times 10^{-13}$

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14. Ans. (A,C)

Sol. There is no strong force between electron & proton. Strong nuclear force becomes repulsive for distance less than 1 fm.

15. Ans. (B,D)

Sol. $\lambda_1 + \lambda_2 + \lambda_3 = \lambda_{eff}$

$$\frac{1}{T_1}+\frac{1}{T_2}+\frac{1}{T_3}=\frac{1}{T}=\frac{1}{10}$$

$$\frac{1}{9}\left[1+\frac{1}{r}+\frac{1}{r^2}\right]=\frac{1}{10}$$

$$\frac{r^2}{70}\left[1+\frac{1}{r}+\frac{1}{r^2}\right]=\frac{1}{10}$$

$(r^2 + r + 1) = 7$

$r^2 + r - 6 = 0$

$(r + 3)(r - 2) = 0$

$r = 2$

$T_1 = a$

$T_2 = ar$

$T_3 = ar^2$

$$a = \frac{70}{r^2}$$

$$a = \frac{70}{(2)^2} = \frac{70}{4} = 17.5$$

$a = 17.5$

$$air = \frac{70+2}{4} = \frac{70}{2} = 3.5$$

16. Ans. (B, D)

Sol. From definition of nuclear fussion and nuclear fission.

17. Ans. (A,B,D)

Sol. Higher binding energy per nucleon means the nucleus is more stable and binding energy is the energy released by nucleus and is always positive

Linked Comprehension Type (1 Para × 3Q.) (1 Para × 2Q.) [3 M (-1)]

(Single Correct Answer Type)

18. Ans. (A)

19. Ans. (C)

20. Ans. (A)

Sol. (18 to 20)

$${}_{84}P_0^{210} \rightarrow {}_{82}Pb^{206} + {}_2He^4$$

Mass converted to energy per reaction is

$$\Delta m = m\left({}_{84}P_0^{210}\right) - \left[m\left({}_{82}Pb^{206}\right) + m\left({}_2He^4\right)\right]$$

$= 5.25$ MeV $= 8.4 \times 10^{-13}$ Joule

Total amount of energy required $= 1.2 \times 10^7$ Joule

Input Energy $= 1.2 \times 10^7 \times 10 = 1.2 \times 10^8$ J

Number of reaction required per day

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$n = \frac{dN}{dt} = \frac{1}{7} \times 10^{21}$ per day

If N is the number of polonium atoms required the

$-\frac{dN}{dt} = \lambda N = n = \lambda N$

$N = \frac{n}{\lambda} = \frac{nT}{0.693} = \frac{200}{7} \times 10^{21}$

Mass of each P_0 – atom = 210 amu

Mass of P_0 required after 693 days = $\left(\frac{200}{7}\right) \times 10^{21} \times 210\text{amu} = 1.0$ gm

Initial mass (m_0) of P_0 $\left(\frac{m}{m_0}\right) = \left(\frac{1}{2}\right)^{t/T}$

$\frac{t}{T} = \frac{693}{138.6} = 5 \Rightarrow m_0 = (2)^5 \times 1g = 32gm$

Initial number of P_0 atoms

$= N_0 = (2)^{t/T}.N = \frac{64}{7} \times 10^{23}$

Initial activity of P_0 :

$R = -\frac{dN}{dt} = \lambda N_0 = \frac{0.693}{T} N_0 = 4.57 \times 10^{21}$ per day

21. Ans. (D)

Sol. KE_{max} of β^{-1}.

$Q = 0.8 \times 10^6$ eV

$KE_P + KE_{\beta^-} + KE_{\bar{\nu}} = Q$

KE_P is almost zero

When $KE_\beta = 0$

Then, $KE_{\bar{\nu}} = Q - KE_P$

$\cong Q$

22. Ans. (C)

Sol. $0 \le KE_{\beta^-} \le Q - KE_P - KE_{\bar{\nu}}$

$0 \le KE_{\beta^-} < Q$

Matching List Type (4 × 4) 1Q.[3 M (–1)]

23. Ans. (C)

Sol. Completing reaction in list II

(1) ${}^{15}_{8}O \rightarrow {}^{15}_{7}N + {}_{1}{}^{0}e$ (β^+ decay Q)

(2) ${}^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}_{2}{}^{4}He$ (Alpha decay P)

(3) ${}^{185}_{83}Bi \rightarrow {}^{184}_{82}Pb + {}_{1}{}^{1}H$ (Proton emission S)

(4) ${}^{239}_{94}Pu \rightarrow {}^{140}_{57}La + ...$ (Fission R)

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SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **4 Q. [4 M (0)]**

24. Ans. 2

Sol. Initially let no. of molecules of A = N_A and that of B = N_B

$$\frac{\lambda N_A}{\lambda N_A + 2\lambda N_B} = \frac{25}{100}$$

$$\therefore\ 100\ \lambda N_A = 25\lambda N_A + 50\lambda N_B \Rightarrow N_A = \frac{2}{3}N_B$$

$$\text{Now, } N_A' = N_A e^{-\lambda t} = \frac{2}{3}N_B e^{-\lambda t}$$

$$N_B' = N_B e^{-2\lambda t}$$

$$\therefore\ \frac{\frac{2}{3}\lambda N_B e^{-\lambda t}}{\frac{2}{3}\lambda N_B e^{-\lambda t} + 2\lambda N_B e^{-2\lambda t}} = \frac{75}{100} = \frac{3}{4}$$

$$\Rightarrow \frac{1}{1+3e^{-\lambda t}} = \frac{3}{4} \Rightarrow 1 = 9e^{-\lambda t} \Rightarrow e^{\lambda t} = 9$$

$$\text{So, } t = \frac{\ell n(9)}{\lambda}\text{hr} = \frac{2\ell n(3)}{\ell n(3)}\text{hr} = 2\text{hr}$$

25. Ans. 6 litre

Sol. $$5 \times \frac{\ell n 2}{15} = \ell n\left(\frac{3.7\times10^4}{296\times v}\right)$$

$$2^{1/3} = \frac{3.7\times10^4}{296\times v}\times 60$$

$v = 6000\ cm^3$

$v = 6$ litre

26. Ans. 8

Sol. $$\left|\frac{dN}{dt}\right| = \lambda N_0 e^{-\lambda t}\ \therefore \ell n\left|\frac{dN}{dt}\right| = \ell n(N_0\lambda) - \lambda t\ \therefore \ell n\left|\frac{dN}{dt}\right| = \ell n(N_0\lambda) - \lambda t \Rightarrow -\lambda = \text{slope} = -\frac{1}{2}\text{year}^{-1} \Rightarrow \lambda = \frac{1}{2}\text{yr}^{-1}$$

$$\therefore t_{1/2} = \frac{0.693}{\lambda} = 1.386$$ given time is 3 times of $t_{1/2}$ $\therefore$ value of p is 8.

27. Ans. 5

Sol. $$A = \frac{A_0}{2^n} \Rightarrow 2^n = \frac{6800}{425}$$; n = No. of half life

$2^n = 16 = 2^4$

$\Rightarrow n = 4$

$\Rightarrow 4 \times t_{1/2} = 10$

$\Rightarrow t_{1/2} = 2.5h$

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SECTION-IV

Matrix Match Type (4 × 5) **1 Q. [8 M (for each entry +2(0)]**

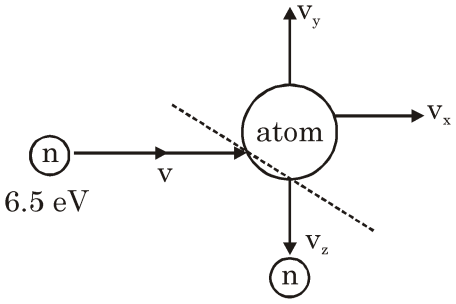
28. Ans. (A) PQ, (B) RST (C) RST (D) RST

Subjective Type **1 Q. [4 M (0)]**

29. Ans. (i) Allowed values of energy of neutron = 6.36 eV and 0.312 eV ; Allowed values of energy of He atom = 17.84 eV and 16.328 eV ,

(ii) 18.23×10^{14} Hz , 9.846×10^{15} Hz , 11.6×10^{15} Hz

Sol. He^+



$m_{He}v_y = m_n v_z \Rightarrow 4v_y = v_z$

$m_n v = m_{He} v_x \qquad v_x = \frac{v}{4}$

$\frac{1}{2}m_n v^2 = 65ev$

$\frac{1}{2}m_n v^2 = \frac{1}{2}m_{He}v_x^2 + \frac{1}{2}m_{He}v_y^2 + \frac{1}{2}m_n v_z^2 + \text{energy in excitation}$

$65ev = \frac{65ev}{4} + \frac{1}{2}4m_0 v_y^2 + \frac{1}{2}m_0 v_z^2 + \text{energy in excitation}$

$65 \times \frac{3}{4} = m_0\left(2v_y^2 + 8v_y^2\right) + \text{energy in excitation}$

$65 \times \frac{3}{4} = 10m_0 v_y^2 + \text{energy in excitation}$

$K.E_{possible}$ for neutron $= (48.75 - 40.8) \times \frac{16}{20} = 6.36$

or $(48.75 - 12.09) \times \frac{16}{20} = 0.312$

$K.E_{atom} = \frac{65}{4} + \frac{6.36}{4}$ or $\frac{65}{4} + \frac{0.312}{4}$

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= 17.84 or 16.328

(ii) The frequency of emitted radiation

40.8 eV = hν or

12.09 × 4 = hν

or 7.56 = hν

$$\nu_1 = \frac{40.8 \times 1.6 \times 10^{-19}}{6.63 \times 10^{-34}} = 9.84 \times 10^{15}\,\text{Hz}$$

$$\nu_2 = \frac{48.36 \times 1.6 \times 10^{-19}}{6.63 \times 10^{-34}} = 11.6 \times 10^{15}\,\text{Hz}$$

$$\nu_3 = \frac{7.56 \times 1.6 \times 10^{-19}}{6.63 \times 10^{-34}} = 18.24 \times 10^{14}\,\text{Hz}$$

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